

01 THE PROBLEM

Every VPN gateway, firewall and bastion must listen on a public address, and anything that listens can be scanned, flooded and exploited before a patch exists. The commercial answer moves control into a vendor cloud you cannot audit or keep sovereign, while security teams still watch anonymous packets and trust logs that root can rewrite.

02 THE INNOVATION

ZeroPort removes the door rather than guarding it. Nothing listens, so a scan returns nothing. Peers prove identity on a threshold-signed directory, meet briefly through a client-authorized rendezvous, then talk over a direct encrypted tunnel. Stronger than port-knocking, which still needs a daemon listening to receive the knock.

03 WHAT ACTUALLY RUNS - NOT A MOCKUP

15/15	BIP-340 Schnorr / secp256k1	against the official specification test vectors
13/13	FROST threshold signatures	2-of-3 emit ONE signature; a lone officer cannot forge it
13/13	Noise IK handshake	WireGuard's own, with forward secrecy and replay window
12/12	NIP-01 over WebSocket	a real Nostr relay; the roster is a signed event
live	OpenTimestamps anchoring	the audit head is committed into Bitcoin
3/3	Real WireGuard kernel interface	peers added and removed by ZeroPort's decisions

04 THE RESULT THAT MATTERS MOST

WireGuard has no access control of its own - it forwards for whoever is in its peer list. ZeroPort supplies the identity, quorum, policy, leases and revocation it lacks. A roster signed by two officers put a real peer into the Windows kernel. A revocation signed by one officer removed it. A roster signed by a single share was rejected, and the kernel never heard about it - a lone administrator is not blocked by a policy check that could be misconfigured; the signature they can produce is not valid at all.

"The kernel now holds a peer that exists only because two officers signed for it."

05 ALIGNMENT - MINISTRY OF INTERIOR & VISION 2030

Directorates, police sites and border posts linked with no public-facing surface. Officer tablets whose access dies with the device. Critical national systems reachable only by enrolled nodes, every access attributed. Seasonal access for Hajj and Umrah, revoked the instant it ends. Relays hosted inside the Kingdom - no foreign coordinator, supporting NCA / ECC controls.

06 STATED LIMITS

No Tor onion routing here; the rendezvous is self-hosted, which for sovereignty is arguably the better answer. Everything runs on one machine, so there is no NAT traversal and no ping figure - one more host closes that. The hosted web page is a mockup of the console; the terminal run is the evidence. A compromised endpoint stays compromised, though its blast radius is bounded.

